

Fixed-Wing Power Simulator — Performance Analysis

Generated 09 April 2026 21:51

Component Weight Summary

Component	Unit (g)	Qty	Total (g)	%
Battery (pack)	300.0	1	300.0	12.0
Motor	120.0	1	120.0	4.8
Airframe / Structure	2080.0	1	2080.0	83.2
TOTAL			2500.0	100.0

Design Inputs

Parameter	Value	Parameter	Value
Weight	2500	Batt Parallel	1
Num Motors	1	Batt Cells S	1
Wing Span	1.6	Batt Cells P	1
Wing Area	0.45	Batt Unit Mode	cell
Cd0	0.028	Motor Kv	920
Cl Max	1.3	Motor I0	0.8
Oswald	0.82	Motor V0	7.0
Mu Roll	0.04	Motor Rated V	16
Cl Takeoff	0.8	Motor R	0.06
Prop Eff	0.75	Motor Imax	40
Cruise Speed	18.0	Motor Pmax	500
Batt Chem	LiPo	Motor Poles	14
Batt Vmin	3.5	Motor Wt	120
Batt Vnom	3.8	Prop D	10
Batt Vmax	4.2	Prop Pitch	4.5
Batt Cell Cap	5000	Prop Blades	2
Batt Cell Wt	75	Prop Maxrpm	0
Batt Chg	5	Prop Maxthr	0
Batt C Cont	25	Altitude	0
Batt C Max	50	Wind	0
Batt Dischg Pct	80	Max V Plot	40
Batt R	5	Periph Cur	0.5
Batt Series	4		

Performance Metrics

Metric	Value	Metric	Value
■ ■ Aircraft ■ ■		■ ■ Battery ■ ■	
Total Weight	2500 g (24.52 N)	Pack Voltage (no load)	16.80 V (1S equiv)
Wing Span	1.600 m	Pack Voltage (under load)	16.76 V
Wing Area	0.4500 m ²	Pack Current	1.77 A
Mean Chord	0.2812 m	Pack Capacity	5000 mAh
Aspect Ratio	5.69	Pack Energy	76.00 Wh
Induced Drag Factor k	0.06824	Usable Energy	60.80 Wh
Lift-curve Slope a	4.649 /rad	■ ■ Propeller ■ ■	
■ ■ Aerodynamics at Cruise ■ ■		Diameter	10.0 in (25.4 cm)
Cruise Airspeed	18.00 m/s (64.8 km/h)	Pitch	4.5 in (11.4 cm)
Stall Speed	8.27 m/s (29.8 km/h)	Estimated RPM	15423
CL at Cruise	0.2745	Tip Speed	205.1 m/s (738.4 km/h)
CD at Cruise	0.03314	Pitch Speed	29.4 m/s (105.8 km/h)
L/D Ratio	8.28	■ ■ Climb Performance ■ ■	
Angle of Attack	3.38 °	Rate of Climb @ Cruise	418.7 m/min (6.98 m/s)
Reynolds Number	346,653	Max Rate of Climb	456.2 m/min @ 23.4 m/s
■ ■ Thrust & Power ■ ■		Max Angle of Climb	24.9 ° @ 11.7 m/s
Thrust Required	2.96 N	Takeoff Ground Roll	23.9 m
Thrust Available	12.46 N	■ ■ Optimal Speeds ■ ■	
Specific Thrust (T/W)	0.508	Best Endurance Speed	9.0 m/s (32.4 km/h)
Power Required (propulsive)	53.3 W	Best Range Speed	11.8 m/s (42.5 km/h)
Motor Electrical Power	19.3 W	Max L/D Ratio	11.44
ESC Losses	0.0 W	■ ■ Endurance & Range @ Cruise ■ ■	
Avionics Power	7.6 W	Flight Time	135.8 min
Total Electrical Power	26.9 W	Flight Range	146.63 km
Max Propeller Power	500.0 W		

Status Checks

Battery Status

Metric	Value	Limit	Notes
Pack voltage (loaded)	16.76 V	≥ 14.00 V	
Pack current vs cont limit	1.8 A	≤ 125.0 A	
Total electrical power	26.9 W	—	
Usable energy	60.8 Wh	—	

Motor / ESC Status

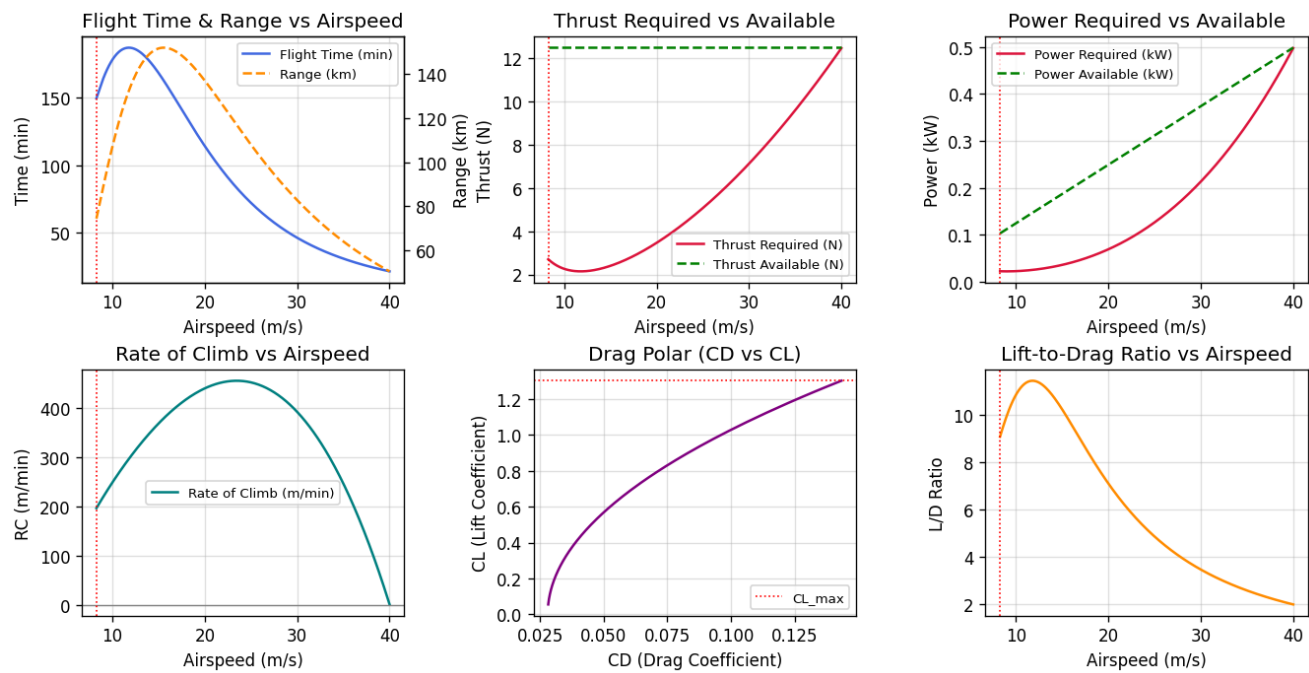
Metric	Value	Limit	Notes
Motor power vs max	19.3 W	≤ 500.0 W	
Thrust Available	12.46 N	—	
Thrust Required	2.96 N	≤ 12.46 N	Must be $< T_{\text{avail}}$ for level flight
Estimated RPM	15423	—	
Prop Tip Speed	205.1 m/s	≤ 200 m/s	Keep below Mach 0.6 (~200 m/s)

Aerodynamic Status

Metric	Value	Limit	Notes
Cruise vs Stall speed	18.0 m/s	> 8.3 m/s	Recommend $> 1.1 \times V_{\text{stall}}$
CL at cruise	0.275	< 1.300	
L/D ratio	8.28	—	
Rate of Climb	418.7 m/min	> 0	

Performance Plots

Fixed-Wing UAV Performance



Simulation Output Log

=== Fixed-Wing Single-Point @ 18.0 m/s (64.8 km/h) ===
Stall Speed : 8.3 m/s
L/D Ratio : 8.28
Thrust Req : 2.96 N | Avail: 12.46 N
Total Power : 26.9 W
Flight Time : 135.8 min
Flight Range : 146.63 km
Rate of Climb : 419 m/min
Max RC : 456 m/min @ 23.4 m/s
Best Endurance: 9.0 m/s | Best Range: 11.8 m/s
Takeoff Roll : 23.9 m